

CO2- ASSESSMENT TOOL - 3

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Question 1

A network administrator has configured a workstation with the following settings:

- * IP Address: 192.168.10.65
- * Subnet Mask: 255.255.255.192 (/26)
- * Default Gateway: 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- * Identifying the subnet to which the configured IP address belongs.
- * Determining the network address, broadcast address, and valid host range.
- * Verifying whether the configured IP address and gateway are valid.
- * Identifying the configuration errors.
- * Optimizing the configuration by assigning an appropriate IP address and default gateway.
- * Calculating the total number of usable host addresses after correction.

IPv4 Network Debugging

Given

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

Step 1: Identify the Subnet A /26 subnet has a block size of 64.

Subnet ranges are:

Subnet	Address Range
192.168.10.0/26	0–63
192.168.10.64/26	64–127
192.168.10.128/26	128–191
192.168.10.192/26	192–255

Since the IP is 192.168.10.65, it belongs to

Subnet: 192.168.10.64/26

Step 2: Calculate Network Details

Network Address

= 192.168.10.64

Broadcast Address

= 192.168.10.127

Host Range

= 192.168.10.65 – 192.168.10.126

Step 3: Verify IP Address

Configured IP

192.168.10.65

It lies inside the host range.

✓ Valid

Step 4: Verify Default Gateway

Gateway configured:

192.168.10.1

This belongs to subnet

192.168.10.0/26

Workstation belongs to

192.168.10.64/26

Hence,

□ Gateway is in a different subnet.

Step 5: Configuration Error

Error:

- Default gateway belongs to another subnet.
- Devices in 192.168.10.64/26 cannot use 192.168.10.1 directly.

Step 6: Optimized Configuration

Parameter	Correct Value
IP Address	192.168.10.65
Subnet Mask	255.255.255.192
Default Gateway	192.168.10.65 cannot be gateway

Assign router interface as 192.168.10.126 or

192.168.10.66

Example:

Gateway = 192.168.10.126

Step 7: Number of Usable Hosts

For /26

$$2^6 = 64$$

Usable Hosts

$$64 - 2 = 62$$

Answer: 62 usable host addresses

Question 2

Two devices are configured with the following IPv6 addresses:

* Device A: 2001:db8::ff00:42:8329/64

* Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- * Expanding both IPv6 addresses into their complete notation.
- * Determining the network prefix of each device.
- * Verifying whether both devices belong to the same subnet.
- * Identifying the addressing or prefix mismatch.
- * Optimizing the IPv6 configuration by assigning appropriate addresses.
- * Calculating the number of available host addresses within the corrected subnet.

IPv6 Debugging

Given

Device A

2001:db8::ff00:42:8329/64

Device B

2001:db8:0:0:1::1/64

Step 1: Expand Addresses

Device A

2001:db8:0000:0000:0000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0001:0000:0000:0001

Step 2: Network Prefix

With /64

First 64 bits are network.

Device A

2001:db8:0000:0000::/64

Device B

2001:db8:0000:0000::/64

Step 3: Same Subnet?

Yes.

Both have identical first four blocks.

✓ Same subnet.

Step 4: Addressing Problem There is no prefix mismatch.

Possible causes:

- Firewall
- Duplicate Address Detection failure
- Incorrect interface configuration
- Disabled IPv6
- Incorrect router advertisement

Step 5: Optimized Configuration

Example

Device A

2001:db8::10/64

Device B

2001:db8::20/64

Gateway

2001:db8::1

Step 6: Number of Hosts

Host bits

$128 - 64 = 64$

Hosts

= 18,446,744,073,709,551,616

Practically unlimited.

Question 3

An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- * Identifying the network address, broadcast address, and host range for each subnet.
- * Calculating the number of usable and unused IP addresses.
- * Identifying inefficient subnet allocations.
- * Debugging the addressing plan by locating subnet allocation mistakes.
- * Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements

IPv4 Subnet Optimization

Network: 192.168.1.0/24

26 Subnet

Item	Value
Network	192.168.1.0
Broadcast	192.168.1.63
Host Range	1–62
Usable Hosts	62

27 Subnet

Item	Value
Network	192.168.1.64
Broadcast	192.168.1.95
Host Range	65–94
Usable Hosts	30

28 Subnet

Item	Value
Network	192.168.1.96
Broadcast	192.168.1.111
Host Range	97–110
Usable Hosts	14

Unused Addresses Example

Suppose Department A requires 20 hosts

Allocated /26

Available

62

Unused

$62 - 20 = 42$

Department B requires 28 hosts

Allocated /27

Available

30

Unused

2

Department C requires 10 hosts

Allocated /28

Available

14

Unused

4

Inefficient Allocation

Department A wastes

42 addresses.

Debugging

Mistakes:

- Large subnet allocated to a small department.
- Future expansion ignored.
- Fixed subnetting wastes addresses.

Optimized VLSM Scheme

Example requirements

Department	Hosts Needed	Recommended Subnet
A	50	/26
B	25	/27
C	10	/28

Benefits

- Better utilization
- Less wastage
- Supports future growth
- Efficient routing

Question 4

Two devices are configured with the following IPv6 addresses:

* Device A: 2001:db8::ff00:42:8329/64

* Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- * Expanding both IPv6 addresses into their complete notation.
- * Determining the network prefix of each device.
- * Verifying whether both devices belong to the same subnet.
- * Identifying the addressing or prefix mismatch.
- * Optimizing the IPv6 configuration by assigning appropriate addresses.
- * Calculating the number of available host addresses within the corrected subnet.

IPv6 Debugging

1. Expanded IPv6 Addresses

- **Device A:** 2001:0db8:0000:0000:0000:ff00:0042:8329/64
- **Device B:** 2001:0db8:0000:0000:0001:0000:0000:0001/64

2. Network Prefix

Since the prefix length is /64, the first 64 bits represent the network.

- **Device A:** 2001:db8:0:0::/64
- **Device B:** 2001:db8:0:0::/64

3. Subnet Verification

Both devices have the same /64 **network prefix**, so they **belong to the same subnet**.

Therefore, there is **no IPv6 prefix mismatch** between the two devices.

4. Configuration Issue

If communication is still failing, the problem may be due to:

- Incorrect default gateway
- Routing or firewall configuration
- VLAN/interface configuration
- Neighbor Discovery Protocol (NDP) failure

5. Optimized Configuration

A simple configuration for the same subnet is:

- **Device A:** 2001:db8:0:0::1/64
- **Device B:** 2001:db8:0:0::2/64

Both belong to **2001:db8:0:0::/64**.

6. Number of Available Addresses

For a /64 IPv6 subnet:

Available addresses = 18,446,744,073,709,551,616

Conclusion

Both devices are correctly placed in the same **/64 subnet**. Hence, the communication failure is likely caused by another configuration issue such as routing, gateway, firewall, VLAN, or NDP.

Question 5

A router contains the following routing table.

Destination Network	Next Hop
---------------------	----------

192.168.1.0/24	10.0.0.2
----------------	----------

192.168.2.0/24	10.0.0.3
----------------	----------

Default Route	10.0.0.1
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Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task

Debug the routing configuration by:

- * Identifying the matching routing entry using the longest-prefix match.
- * Determining the next-hop router selected for packet forwarding.
- * Verifying whether the destination IP belongs to the intended subnet.
- * Identifying any routing configuration errors.
- * Optimizing the routing table to improve packet forwarding efficiency.
- * Identifying the route that will be used if no matching destination exist

Routing Table Debugging

Routing Table

Destination	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default	10.0.0.1

Destination Packet

192.168.2.100

Step 1: Longest Prefix Match

Destination

192.168.2.100

Matches

192.168.2.0/24

✓ Longest Prefix Match

Step 2: Next Hop

Next Hop

10.0.0.3

Step 3: Verify Destination

Subnet

192.168.2.0/24

Host Range

192.168.2.1 – 192.168.2.254

Destination

192.168.2.100

✓ Valid

Step 4: Routing Errors Possible issues

- Incorrect next-hop IP
- Down router interface

- Missing return route
- ARP failure
- Interface shutdown

Step 5: Optimized Routing

Destination	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
192.168.3.0/24	10.0.0.4 (if required)
Default	10.0.0.1

Improvements

- Verify reachability of next hops.
- Remove unused routes.
- Add missing routes where needed.
- Use route summarization if multiple contiguous networks exist.

Step 6: If No Matching Route Exists

The router forwards the packet using the

Default Route → 10.0.0.1

Final Answers Summary

Question	Final Answer
Q1	IP belongs to 192.168.10.64/26, gateway is incorrect, correct gateway should be in 192.168.10.64/26, usable hosts = 62.
Q2	Both devices belong to 2001:db8:0:0::/64; no prefix mismatch; corrected example addresses: 2001:db8::10/64, 2001:db8::20/64; available hosts = 2^{64} .
Q3	Fixed subnetting wastes addresses; optimize using VLSM to allocate /26, /27, and /28 according to department sizes.
Q4	Same as Q2: both devices are on 2001:db8:0:0::/64; no addressing mismatch; available hosts = 2^{64} .
Q5	Destination 192.168.2.100 matches 192.168.2.0/24; next hop is 10.0.0.3; if no route matches, the default route (10.0.0.1) is used.